

Οικονομικό Πανεπιστήμιο Αθηνών
Σχολή Διοίκησης Επιχειρήσεων
Τμήμα Διοικητικής Επιστήμης και Τεχνολογίας
Τεχνολογίες Λογισμικού στη Πράξη



6^ο Εξάμηνο

Lab Assignment
Γεωργία Σεϊντή (8230132)

Διδάσκων Καθηγητής: κ. Διομήδης Σπινέλλης

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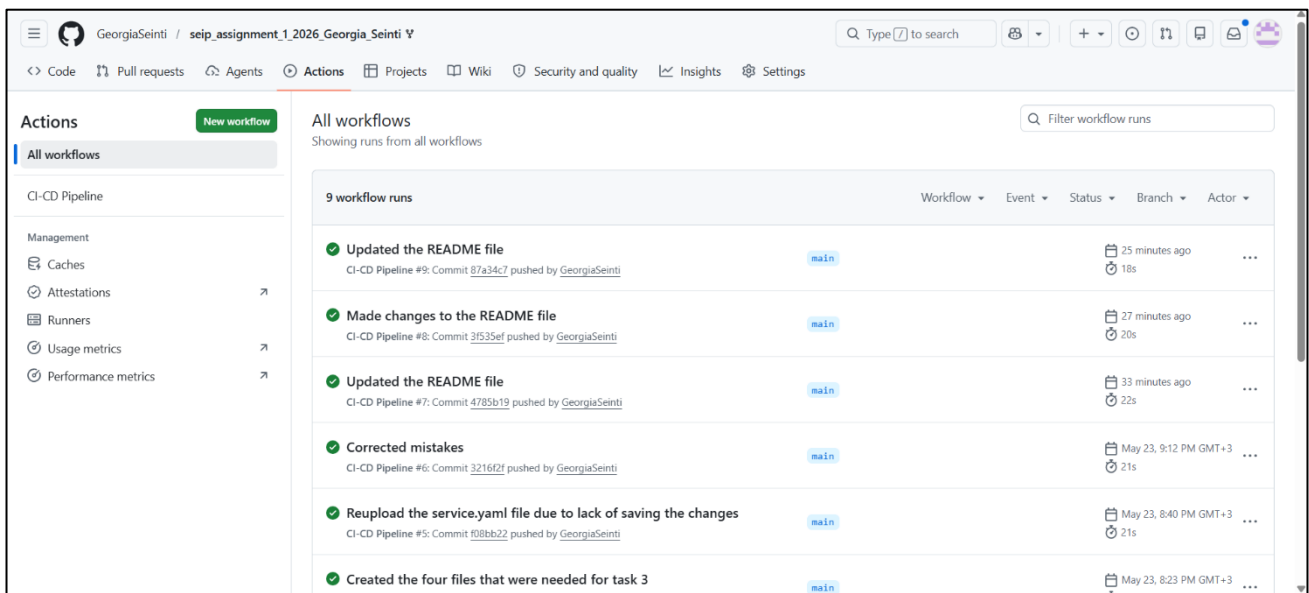
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1. GitHub Repository Link

You can find the contents of my fork in the following link:

https://github.com/GeorgiaSeinti/seip_assignment_1_2026_Georgia_Seinti

2. CI/CD Proof



3. Cluster State Proof

The following image shows the clean output of the command `kubectl get all -n default`. It shows 3 healthy running pods, the deployment status and the ClusterIP service.

```

PS C:\Users\georg\OneDrive\Υπολογιστής\spin lab\seip_assignment_1_2026_Georgia_Seinti> kubectl get all -n default
NAME                                     READY   STATUS    RESTARTS   AGE
pod/echo-api-deployment-7c4877cf5c-9nvbb 1/1     Running   0           6m41s
pod/echo-api-deployment-7c4877cf5c-j6k94 1/1     Running   0           6m41s
pod/echo-api-deployment-7c4877cf5c-tmktt 1/1     Running   0           6m41s

NAME                                TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
service/echo-api-service            ClusterIP     10.99.17.157 <none>        80/TCP     9m34s
service/kubernetes                  ClusterIP     10.96.0.1    <none>        443/TCP    22m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/echo-api-deployment 3/3     3             3           6m42s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/echo-api-deployment-7c4877cf5c 3         3         3       6m42s
PS C:\Users\georg\OneDrive\Υπολογιστής\spin lab\seip_assignment_1_2026_Georgia_Seinti>

```

In this picture is shown the clean output of the command `kubectl get configmap,secret`.

```

PS C:\Users\georg\OneDrive\Υπολογιστής\spin lab\seip_assignment_1_2026_Georgia_Seinti> kubectl get configmap,secret
NAME                                DATA   AGE
configmap/echo-api-config           2       11m
configmap/kube-root-ca.crt          1       23m

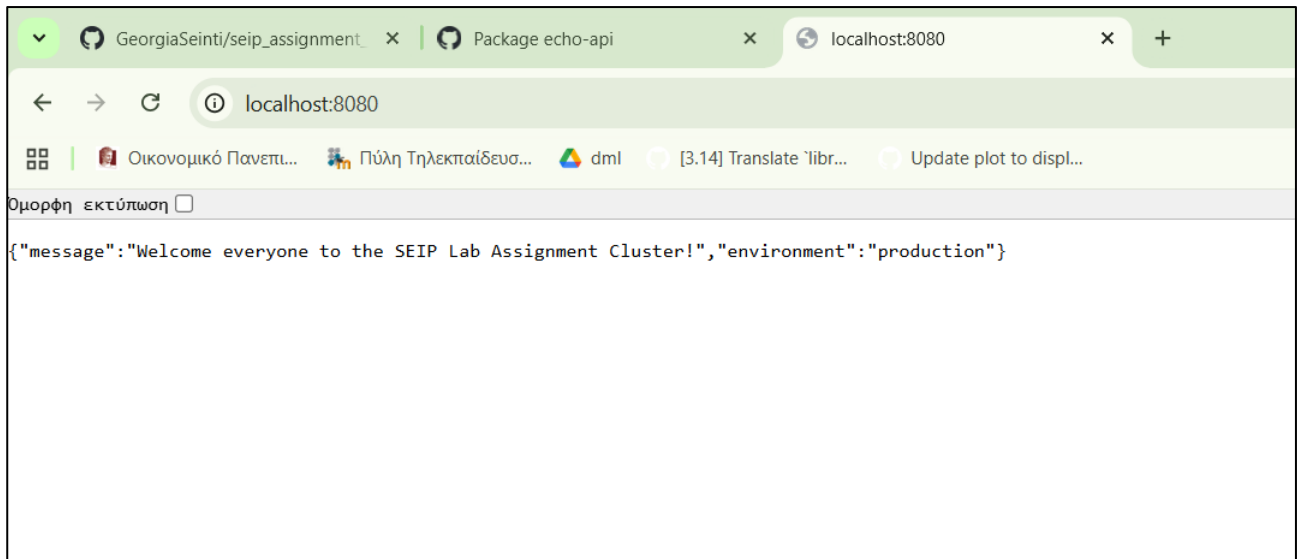
NAME                                TYPE      DATA   AGE
secret/echo-api-secret              Opaque    1       10m
PS C:\Users\georg\OneDrive\Υπολογιστής\spin lab\seip_assignment_1_2026_Georgia_Seinti>

```

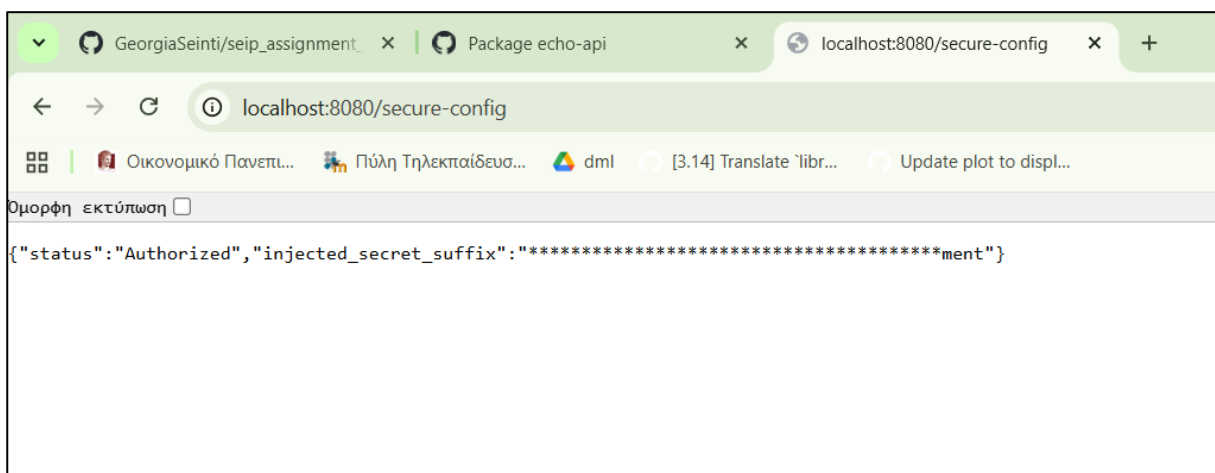
4. Application Verification Proof

For the application verification proof, I used a browser engine and accessed the application through the local development URL: <http://localhost:8080>

The application successfully displayed the custom ConfigMap greeting message:
 “Welcome everyone to the SEIP Lab Assignment Cluster!”



Then, by accessing the application through the local development URL: <http://localhost:8080/secure-config>, the application successfully displayed the status “Authorized” along with the masked string suffix of the configured secret. In order to encode the secret, I used the following command: `echo -n "georgina_secret_key_for_seip_lab_assignment" | base64`



5. AI Reflection & Future Outlook

5.1. AI Integration

For this assignment, I used Generative AI tools and to be more specific ChatGPT and Claude for two main reasons. The first one was to better understand what Kubernetes and Minicube are, how they work and what the commands that we had to run according to the assignment guidelines do. The second one is to familiarize myself with the syntax of Dockerfiles and the YAML files. Since I had not previously worked with these types of files beyond a theoretical level, I needed some guidance on how they are structured and written in real world use cases. On a smaller scale, I also used these tools to help interpret error messages that appeared in the terminal, as well as to improve the quality of my English both in the README file and in this report.

5.2. Utility Analysis

In my opinion, the most useful aspect of AI assistance was the understanding of the syntax of Dockerfiles and YAML files. It helped me grasp how they work in practice and how I can apply the requirements outlined in the assignment guidelines within these files. Another aspect in which AI assistance was proved useful is for debugging. I have two examples in mind. The first is a problem that I had in the ci-cd.yaml file. When I was building and tagging the image I wrote ghcr.io/<GeorgiaSeinti>/echo-api:latest, because I thought that we were supposed to write our GitHub username inside the brackets. This, of course, led to an error and AI helped me understand how to correct it. Another example of how AI helped me solve a problem was an issue that occurred in the GitHub Actions workflow. After a certain point, even though the workflow had run successfully before, it started failing. By investigating the workflow run logs, I found where the error is and with the help of AI, I realized that I had to modify the Workflow permissions inside the GitHub Actions settings and set them to “Read and write permissions”.

5.3. Friction Points

I did not encounter any major friction points. However, there was an instance in which AI tools provided incorrect information. It concerned the

problem I mentioned in the previous question regarding the name of the image. In the beginning, AI suggested that it was sufficient to remove the brackets and that there was no need to write the whole of the username in lowercase. Although this was not a significant issue, I had to conduct additional research in order to determine the correct formatting of the username.

5.4. Future Architectural Outlook

If I had an extra week or if I was tasked with scaling this pipeline for a real-world enterprise system I would focus on enhancing scalability, observability and deployment automation. To begin with, to manage routing in a more production-ready way, I would replace the port-forward approach with a Kubernetes Ingress Controller. This would facilitate traffic routing and domain management.

To improve monitoring and operational visibility, I would integrate Prometheus, a monitoring-alerting system for monitoring containerized or traditional applications that collects and stores real-time metrics. This way, we can have a collection of metrics from both the application and the Kubernetes cluster. As a result, we have proactive monitoring of resource use, performance of the system and application health.

Finally, I would implement a GitOps deployment workflow using ArgoCD, which is a continuous delivery tool for Kubernetes that automatically synchronizes the cluster state with the manifests stored in Git. It provides automated tracking, version control and in general greater deployment reliability.